CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · NEUROLOGY PHASE 2,3 · STARTS 2015–2024

Dopaminergic: 3 molecules stopped early, all small molecules

4 stopped trials, but 3 molecules — all small molecules: Tavapadon, Loxapine and Lumateperone. Against 9.6% for all neurology trials with a resolved drug target and 6.4% across all neurology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

7.7%
4 of 52 closed trials (95% CI 3.0%–18.2%)

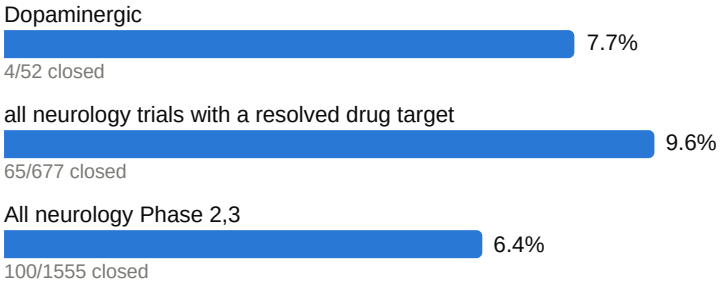
3 / 1 / 0
efficacy / safety / both (adds to 4)

72.2%
of 72 trials have closed · 20 still open or unresolved

6.8%
stopped for a biological reason within 3 years of starting (95% CI 2.6%–17.1%; 20 still at risk; 3.4% area-wide)

1 of 12 **Completed and missed the primary endpoint.** Of 45 completed trials in this segment, 12 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 1 missed and 11 met. Counted separately and never in the rate above; a floor, not a rate. 11 of the 43 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 133 distinct experimental drugs across 72 trials (small molecule 72, unknown 13, protein 1). Most tested: LEVODOPA (10), APOMORPHINE (9), RASAGILINE (6), TAVAPADON (6), CARBIDOPA ANHYDROUS (5). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 4 stops came from 3 sponsor-asset programmes across 3 sponsors; the largest (Pfizer) contributed 2; removing that programme's trials from both sides leaves 4.0%. Screened alongside 27 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 2 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 11.5%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	4	28	14.3%	5.7%–31.5%
2018–2020	0	14	0.0%	0.0%–21.5%
2021–2024	0	10	0.0%	0.0%–27.8%

Where the stops sit

Efficacy / futility: 3, Safety: 1.
Sponsors with most stops: Pfizer (2), Intra-Cellular Therapies, Inc. (1), University of Witten/Herdecke (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT02817906	3	Intra-Cellular Therapies, Inc.	LUMATEPERONE	2019-01	Efficacy	Pre-specified Interim Analysis Indicated Futility
NCT02847650	2	Pfizer PFE	TAVAPADON	2018-01	Efficacy	Study B7601011 was terminated on 29 Jan 2018 due to lack of efficacy in moderate/advanced Parkinson's disease.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT02687542	2	Pfizer PFE	TAVAPADON, PF-06649751 middle dose 1 (3 mg Q…, PF-06649751 middle dose 2 (7 mg Q…	2017-11	Efficacy	Terminated 25Sep17 due to insufficient efficacy. Not due to safety reasons.
NCT02820519	2	University of Witten/Herdecke	LOXAPINE	2017-05	Safety	Intolerable high amount of adverse events

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 neurology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 45% of industry neurology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 72.2% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.